

Limits and Continuity

Real Analysis

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Definition

Let $a \in \mathbb{R}$ and $\varepsilon > 0$. Then the ε -neighborhood of a is the set

$$V_\varepsilon(a) := \{x \in \mathbb{R} : |x - a| < \varepsilon\}.$$

Definition

Let $A \subseteq \mathbb{R}$. A point $c \in \mathbb{R}$ is a **cluster point** (or **accumulation point**) of A if for every $\delta > 0$ there exists at least one point $x \in A$, $x \neq c$, such that $|x - c| < \delta$.

Definition

Let $A \subseteq \mathbb{R}$ and let c be a cluster point of A . For a function $f : A \rightarrow \mathbb{R}$, a real number L is said to be the **limit of f at c** if, given any $\varepsilon > 0$, there exists $\delta > 0$ such that if $x \in A$ and $0 < |x - c| < \delta$, then

$$|f(x) - L| < \varepsilon.$$

Remark

If L is the limit of f at c , then we also say that f **converges to L at c** . We frequently write

$$L = \lim_{x \rightarrow c} f(x) \quad \text{or} \quad L = \lim_{x \rightarrow c} f.$$

We also sometimes use the notation

$$f(x) \rightarrow L \quad \text{as} \quad x \rightarrow c$$

to express the fact that f has limit L at c .

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Theorem

Let $f : A \rightarrow \mathbb{R}$ and let c be a cluster point of A . Then the following statements are equivalent:

- i) $\lim_{x \rightarrow c} f(x) = L$.
- ii) *Given any ε -neighborhood $V_\varepsilon(L)$ of L , there exists a δ -neighborhood $V_\delta(c)$ of c such that if $x \neq c$ is any point in $V_\delta(c) \cap A$, then $f(x)$ belongs to $V_\varepsilon(L)$.*

Definition

Let $A \subseteq \mathbb{R}$, let $f : A \rightarrow \mathbb{R}$, and let $c \in \mathbb{R}$ be a cluster point of A . We say that f is **bounded on a neighborhood of c** if there exists a δ -neighborhood $V_\delta(c)$ of c and a constant $M > 0$ such that $|f(x)| \leq M$ for all $x \in V_\delta(c) \cap A$.

Theorem

If $A \subseteq \mathbb{R}$ and $f : A \rightarrow \mathbb{R}$ has a limit at $c \in \mathbb{R}$, then f is bounded on some neighborhood of c .

Definition

Let $A \subseteq \mathbb{R}$ and let f and g be functions from A to $\mathbb{R}$. We define the **sum** $f+g$, the **difference** $f-g$, and the **product** fg from A to $\mathbb{R}$ as the functions given by

$$(f+g)(x) := f(x)+g(x), \quad (f-g)(x) := f(x)-g(x), \quad (fg)(x) := f(x)g(x),$$

for all $x \in A$. Moreover, if $b \in \mathbb{R}$, we define the **multiple** bf as the function given by

$$(bf)(x) := bf(x) \quad \text{for all } x \in A.$$

Finally, if $h(x) \neq 0$ for $x \in A$, we define the **quotient** f/h as the function given by

$$\left(\frac{f}{h}\right)(x) := \frac{f(x)}{h(x)} \quad \text{for all } x \in A.$$

Theorem

Let $A \subseteq \mathbb{R}$, let f and g be functions from A to $\mathbb{R}$, and let $c \in \mathbb{R}$ be a cluster point of A . Moreover, let $b \in \mathbb{R}$.

a) If $\lim_{x \rightarrow c} f(x) = L$ and $\lim_{x \rightarrow c} g(x) = M$, then:

$$\lim_{x \rightarrow c} (f + g) = L + M, \quad \lim_{x \rightarrow c} (f - g) = L - M,$$

$$\lim_{x \rightarrow c} (fg) = LM, \quad \lim_{x \rightarrow c} (bf) = bL.$$

b) If $h : A \rightarrow \mathbb{R}$, if $h(x) \neq 0$ for all $x \in A$, and if $\lim_{x \rightarrow c} h = H \neq 0$, then

$$\lim_{x \rightarrow c} \left(\frac{f}{h} \right) = \frac{L}{H}.$$

Theorem

Let $A \subseteq \mathbb{R}$, let $f : A \rightarrow \mathbb{R}$, and let $c \in \mathbb{R}$ be a cluster point of A . If

$$a \leq f(x) \leq b \quad \text{for all } x \in A, x \neq c,$$

and if $\lim_{x \rightarrow c} f$ exists, then $a \leq \lim_{x \rightarrow c} f \leq b$.

Theorem (Squeeze Theorem)

Let $A \subseteq \mathbb{R}$, let $f, g, h : A \rightarrow \mathbb{R}$, and let $c \in \mathbb{R}$ be a cluster point of A . If

$$f(x) \leq g(x) \leq h(x) \quad \text{for all } x \in A, x \neq c,$$

and if $\lim_{x \rightarrow c} f = L = \lim_{x \rightarrow c} h$, then $\lim_{x \rightarrow c} g = L$.

Theorem

Let $A \subseteq \mathbb{R}$, let $f : A \rightarrow \mathbb{R}$, and let $c \in \mathbb{R}$ be a cluster point of A . If

$$\lim_{x \rightarrow c} f > 0 \quad \left[\text{respectively, } \lim_{x \rightarrow c} f < 0 \right],$$

then there exists a δ -neighborhood $V_\delta(c)$ of c such that $f(x) > 0$ [respectively, $f(x) < 0$] for all $x \in V_\delta(c) \cap A$, $x \neq c$.

Definition

Let $A \subseteq \mathbb{R}$ and let $f : A \rightarrow \mathbb{R}$.

- i) If $c \in \mathbb{R}$ is a cluster point of the set $A \cap (c, \infty) = \{x \in A : x > c\}$, then $L \in \mathbb{R}$ is said to be the **right-hand limit of f at c** , written

$$\lim_{x \rightarrow c^+} f(x) = L \quad \text{or} \quad \lim_{x \rightarrow c^+} f = L,$$

if given any $\varepsilon > 0$ there exists $\delta = \delta(\varepsilon) > 0$ such that for all $x \in A$ with $0 < x - c < \delta$, then $|f(x) - L| < \varepsilon$.

- ii) If $c \in \mathbb{R}$ is a cluster point of the set $A \cap (-\infty, c) = \{x \in A : x < c\}$, then $L \in \mathbb{R}$ is said to be the **left-hand limit of f at c** , written

$$\lim_{x \rightarrow c^-} f(x) = L \quad \text{or} \quad \lim_{x \rightarrow c^-} f = L,$$

if given any $\varepsilon > 0$ there exists $\delta = \delta(\varepsilon) > 0$ such that for all $x \in A$ with $0 < c - x < \delta$, then $|f(x) - L| < \varepsilon$.

Theorem

Let $A \subseteq \mathbb{R}$, let $f : A \rightarrow \mathbb{R}$, and let $c \in \mathbb{R}$ be a cluster point of both $A \cap (c, \infty)$ and $A \cap (-\infty, c)$. Then $\lim_{x \rightarrow c} f = L$ if and only if

$$\lim_{x \rightarrow c^+} f = L = \lim_{x \rightarrow c^-} f.$$

Definition

Let $A \subseteq \mathbb{R}$, let $f : A \rightarrow \mathbb{R}$, and let $c \in \mathbb{R}$ be a cluster point of A .

i) We say that f **tends to** ∞ **as** $x \rightarrow c$, written

$$\lim_{x \rightarrow c} f = \infty,$$

if for every $\alpha \in \mathbb{R}$ there exists $\delta = \delta(\alpha) > 0$ such that for all $x \in A$ with $0 < |x - c| < \delta$, then $f(x) > \alpha$.

ii) We say that f **tends to** $-\infty$ **as** $x \rightarrow c$, written

$$\lim_{x \rightarrow c} f = -\infty,$$

if for every $\beta \in \mathbb{R}$ there exists $\delta = \delta(\beta) > 0$ such that for all $x \in A$ with $0 < |x - c| < \delta$, then $f(x) < \beta$.

Theorem

Let $A \subseteq \mathbb{R}$, let $f, g : A \rightarrow \mathbb{R}$, and let $c \in \mathbb{R}$ be a cluster point of A . Suppose that $f(x) \leq g(x)$ for all $x \in A$, $x \neq c$.

- a) If $\lim_{x \rightarrow c} f = \infty$, then $\lim_{x \rightarrow c} g = \infty$.
- b) If $\lim_{x \rightarrow c} g = -\infty$, then $\lim_{x \rightarrow c} f = -\infty$.

Definition

Let $A \subseteq \mathbb{R}$ and let $f : A \rightarrow \mathbb{R}$. If $c \in \mathbb{R}$ is a cluster point of the set $A \cap (c, \infty)$, then we say that f **tends to** ∞ [respectively, to $-\infty$] as $x \rightarrow c^+$, written

$$\lim_{x \rightarrow c^+} f = \infty \quad \left[\text{respectively, } \lim_{x \rightarrow c^+} f = -\infty \right],$$

if for every $\alpha \in \mathbb{R}$ there exists $\delta = \delta(\alpha) > 0$ such that for all $x \in A$ with $0 < x - c < \delta$, then $f(x) > \alpha$ [respectively, $f(x) < \alpha$].

Definition

Let $A \subseteq \mathbb{R}$ and let $f : A \rightarrow \mathbb{R}$. Suppose that $(a, \infty) \subseteq A$ for some $a \in \mathbb{R}$. We say that $L \in \mathbb{R}$ is the **limit of f as $x \rightarrow \infty$** , written

$$\lim_{x \rightarrow \infty} f(x) = L \quad \text{or} \quad \lim_{x \rightarrow \infty} f = L,$$

if given any $\varepsilon > 0$ there exists $K = K(\varepsilon) > a$ such that for any $x > K$, then $|f(x) - L| < \varepsilon$.

Definition

Let $A \subseteq \mathbb{R}$ and let $f : A \rightarrow \mathbb{R}$. Suppose that $(a, \infty) \subseteq A$ for some $a \in \mathbb{R}$. We say that f **tends to** ∞ [respectively, to $-\infty$] as $x \rightarrow \infty$, written

$$\lim_{x \rightarrow \infty} f = \infty \quad \left[\text{respectively, } \lim_{x \rightarrow \infty} f = -\infty \right],$$

if given any $\alpha \in \mathbb{R}$ there exists $K = K(\alpha) > a$ such that for any $x > K$, then $f(x) > \alpha$ [respectively, $f(x) < \alpha$].

Theorem

Let $A \subseteq \mathbb{R}$, let $f, g : A \rightarrow \mathbb{R}$, and suppose that $(a, \infty) \subseteq A$ for some $a \in \mathbb{R}$. Suppose also that $g(x) > 0$ for all $x > a$ and that for some $L \in \mathbb{R}$, $L \neq 0$, we have

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = L.$$

- i) If $L > 0$, then $\lim_{x \rightarrow \infty} f = \infty$ if and only if $\lim_{x \rightarrow \infty} g = \infty$.
- ii) If $L < 0$, then $\lim_{x \rightarrow \infty} f = -\infty$ if and only if $\lim_{x \rightarrow \infty} g = \infty$.

Definition

Let $A \subseteq \mathbb{R}$, let $f : A \rightarrow \mathbb{R}$, and let $c \in A$. We say that f **is continuous at c** if, given any $\varepsilon > 0$, there exists $\delta > 0$ such that if x is any point of A satisfying $|x - c| < \delta$, then $|f(x) - f(c)| < \varepsilon$. If f is not continuous at c , then we say that f **is discontinuous at c** .

Theorem

A function $f : A \rightarrow \mathbb{R}$ is continuous at a point $c \in A$ if and only if given any ε -neighborhood $V_\varepsilon(f(c))$ of $f(c)$ there exists a δ -neighborhood $V_\delta(c)$ of c such that if x is any point of $V_\delta(c) \cap A$, then $f(x)$ belongs to $V_\varepsilon(f(c))$, that is,

$$f(V_\delta(c) \cap A) \subseteq V_\varepsilon(f(c)).$$

Definition

*Let $A \subseteq \mathbb{R}$ and let $f : A \rightarrow \mathbb{R}$. If B is a subset of A , we say that f is **continuous on the set B** if f is continuous at each point of B .*

Theorem

Let $A \subseteq \mathbb{R}$, let f and g be functions from A to $\mathbb{R}$, and let $b \in \mathbb{R}$. Suppose that $c \in A$ and that f and g are continuous at c .

- a) Then $f + g$, $f - g$, fg , and bf are continuous at c .*
- b) If $h : A \rightarrow \mathbb{R}$ is continuous at $c \in A$ and if $h(x) \neq 0$ for all $x \in A$, then the quotient f/h is continuous at c .*

Theorem

Let $A \subseteq \mathbb{R}$, let f and g be continuous functions from A to $\mathbb{R}$, and let $b \in \mathbb{R}$.

- a) The functions $f + g$, $f - g$, fg , and bf are continuous on A .*
- b) If $h : A \rightarrow \mathbb{R}$ is continuous on A and if $h(x) \neq 0$ for $x \in A$, then the quotient f/h is continuous on A .*

Theorem

Let $A \subseteq \mathbb{R}$, let $f : A \rightarrow \mathbb{R}$, and let $|f|$ be defined by $|f|(x) := |f(x)|$ for $x \in A$.

- a) If f is continuous at a point $c \in A$, then $|f|$ is continuous at c .*
- b) If f is continuous on A , then $|f|$ is continuous on A .*

Theorem

Let $A \subseteq \mathbb{R}$, let $f : A \rightarrow \mathbb{R}$, and let $f(x) \geq 0$ for all $x \in A$. We define $\sqrt{f}$ for $x \in A$ by $(\sqrt{f})(x) := \sqrt{f(x)}$.

- a) If f is continuous at a point $c \in A$, then $\sqrt{f}$ is continuous at c .
- b) If f is continuous on A , then $\sqrt{f}$ is continuous on A .

Theorem

Let $A, B \subseteq \mathbb{R}$ and let $f : A \rightarrow \mathbb{R}$ and $g : B \rightarrow \mathbb{R}$ be functions such that $f(A) \subseteq B$. If f is continuous at a point $c \in A$ and g is continuous at $b = f(c) \in B$, then the composition $g \circ f : A \rightarrow \mathbb{R}$ is continuous at c .

Theorem

Let $A, B \subseteq \mathbb{R}$, let $f : A \rightarrow \mathbb{R}$ be continuous on A , and let $g : B \rightarrow \mathbb{R}$ be continuous on B . If $f(A) \subseteq B$, then the composite function $g \circ f : A \rightarrow \mathbb{R}$ is continuous on A .

Definition

A function $f : A \rightarrow \mathbb{R}$ is said to be **bounded on** A if there exists a constant $M > 0$ such that $|f(x)| \leq M$ for all $x \in A$.

Theorem (Boundedness Theorem)

Let $I := [a, b]$ be a closed bounded interval and let $f : I \rightarrow \mathbb{R}$ be continuous on I . Then f is bounded on I .

Definition

Let $A \subseteq \mathbb{R}$ and let $f : A \rightarrow \mathbb{R}$. We say that f **has an absolute maximum on A** if there exists a point $x^* \in A$ such that

$$f(x^*) \geq f(x) \quad \text{for all } x \in A.$$

We say that f **has an absolute minimum on A** if there exists a point $x_* \in A$ such that

$$f(x_*) \leq f(x) \quad \text{for all } x \in A.$$

We say that x^* is an **absolute maximum point of f on A** and that x_* is an **absolute minimum point of f on A** , if they exist.

Theorem (Maximum-Minimum Theorem)

Let $I := [a, b]$ be a closed bounded interval and let $f : I \rightarrow \mathbb{R}$ be continuous on I . Then f has an absolute maximum and an absolute minimum on I .

Theorem (Location of Roots Theorem)

Let $I := [a, b]$ and let $f : I \rightarrow \mathbb{R}$ be continuous on I . If $f(a) < 0 < f(b)$, or if $f(a) > 0 > f(b)$, then there exists a number $c \in (a, b)$ such that $f(c) = 0$.

Theorem (Bolzano's Intermediate Value Theorem)

Let I be an interval and let $f : I \rightarrow \mathbb{R}$ be continuous on I . If $a, b \in I$ and if $k \in \mathbb{R}$ satisfies $f(a) < k < f(b)$, then there exists a point $c \in I$ between a and b such that $f(c) = k$.

Corollary

Let $I := [a, b]$ be a closed bounded interval and let $f : I \rightarrow \mathbb{R}$ be continuous on I . If $k \in \mathbb{R}$ is any number that satisfies

$$\inf f(I) \leq k \leq \sup f(I),$$

then there exists a number $c \in I$ such that $f(c) = k$.

Theorem

Let I be a closed bounded interval and let $f : I \rightarrow \mathbb{R}$ be continuous on I . Then the set $f(I) := \{f(x) : x \in I\}$ is a closed bounded interval.

Theorem (Preservation of Intervals Theorem)

Let I be an interval and let $f : I \rightarrow \mathbb{R}$ be continuous on I . Then the set $f(I)$ is an interval.